

Effect of Quantum Spacetime Fluctuations on the Stability of White Dwarf

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Introduction

Generalized Uncertainty Principle and Non-commutative Geometry

- Theories of quantum gravity, such as black-hole physics, string theory, and loop quantum gravity, predict the existence of a fundamental length scale¹
 - Micro Black Hole - Gedanken experiment.
 - String Theory - The string cannot interact with distance smaller than its size.
 - Loop Quantum Gravity - The existence of bounce in LQC (Loop Quantum Cosmology).
- This minimal length scale can be incorporated into quantum mechanics by generalising the uncertainty relation. A relation such as $\Delta x \Delta p \geq \frac{\hbar}{2} \left[1 + \beta (\Delta p)^2 + \beta \langle p \rangle^2 \right]$ admits a minimum position uncertainty², $\Delta x_{\min} = \hbar \sqrt{\beta}$.
- To generate the above, the operators $\hat{x}_i$ and $\hat{p}_j$ obey a relation $[\hat{x}_i, \hat{p}_j] = i\hbar(\delta_{ij} + \beta p_i p_j)$.
- If we require $[p_i, p_j] = 0$ then the commutation relations among the position operators lead to a non-commutative geometry such as $[x_i, x_j] = 2i\hbar\beta(\hat{p}_i \hat{x}_j - \hat{p}_j \hat{x}_i)$.
- In this consideration the original position operator are no longer self-adjoint and cannot form a physical state with position basis.
- But one can retrieve the information about the position by constructing maximal localization states. And these state has certain fuzziness associated with it.
- On the other hand, Non-commutative geometry corresponds to the algebra³ where space-time coordinates follows $[x^\mu, x^\nu] = i\theta^{\mu\nu}$ with $\theta^{\mu\nu}$ is an antisymmetric matrix.

¹D. Amati, M. Ciafaloni, and G. Veneziano, Phys. Lett. B 216, 41 (1989), K. Konishi, G. Pauti, and P. Provero, Phys. Lett. B 234, 276 (1990), M. Maggiore, Phys. Rev. D 49, 5182 (1994), T. Padmanabhan, Gen. Rel. Gravit. 17, 215 (1985), J. Greensite, Phys. Lett. B 255, 375 (1991).

²A. Kempf, G. Mangano, and R. B. Mann, Phys. Rev. D 52, 1108 (1995)

³S. H.S. Alexander, J. Magueijo, arXiv:hep-th/0104093.

Significance

Generalized Uncertainty Principle and Non-commutative Geometry

- Non-commutative geometry was initially observed as an outcome of a discrete Lorentz invariant space-time to address the problem of divergence in quantum field theory⁴.
- But it was later argued by Seiberg and Witten⁵ that quantum effect of gravity at Planck scale lead to non-commutative space-time.
- Non-commutativity of space-time are unavoidable even in Minkowski vacuum. It give a foamy structure⁶ to spacetime at Planckian scale.

Interesting points

- The commutation relations $[x_i, x_j] = [p_i, p_j] = 0$ and $[x_i, p_j] = i\hbar \left\{ \delta_{ij} - \lambda \frac{\ell_P}{\hbar} \left(p \delta_{ij} + \frac{p_i p_j}{p} \right) \right\}$ predict a negative deceleration parameter in the early universe and in the late universe⁷ when an additional dilaton field is considered in Friedmann-Robertson-Walker metric.
- In another work⁸, a quantum bounce was observed when $[x, p] = i\hbar \{1 - 2\alpha p\}$ was studied in Friedmann cosmology. Moreover, the resulting Friedmann equations have features similar to those predicted by Loop Quantum Cosmology.

⁴ H. S. Snyder, Phys. Rev. 71, 38 (1947), — Phys. Rev. 72, 68 (1947).

⁵ J. High Energy Phys. 09 032 (1999).

⁶ G. Amelino-Camelio, S. Majid, Int. J. Mod. Phys. A 15, 4301 (2000).

⁷ Majumder2011

⁸

Compact Stars and Quantum Spacetime Fluctuations

- Einstein's field equations are written for commutative space-time coordinates; constructing Einstein's field equations in non-commutative space-time is non-trivial.
- So, we study the behaviour of matter in flat non-commutative space-time which give rise to a modified dispersion relation and then we construct the corresponding energy momentum tensor.
- This yields some perspective regards the behaviour of matter in the quantised spacetime.

Noncommutative geometry

Generalized Uncertainty relation

- To calculate thermodynamic quantities, we need to know the corresponding invariant phase space measure since the relation connecting phase coordinates (x, p) are now modified.
- The analog Liouville theorem suggest^a the invariant measure as $\frac{d^3x}{(2\pi\hbar)^3} \frac{d^3p}{(1+\beta p^2)^3}$.
- In this case number of state transforms as $\sum_p \equiv \int \frac{V d^3p}{h^3} \frac{1}{(1+\beta p^2)^3}$.
- The GUP deformation factor $(1 + \beta p^2)^{-3}$ imposes restriction on the allowed momentum states.
or in other words it force an effective cutoff $p_{\max} \sim \beta^{-1/2}$ in the higher momentum states.

- We examine the effect of noncommutative geometry on white dwarfs for the following algebra^a $[x^i, t] = i\lambda x^i$ and $[x^i, x^j] = 0$ where λ is constant and $i, j = 1, 2, 3$.
- The above noncommutative geometry suggests a modified dispersion relation $E_p^2 = p^2 c^2 (1 + \lambda E_p)^2 + m^2 c^4$.
- Interestingly, this dispersion relation admits a maximum momentum cut-off $p_{\max} = 1/\lambda c$.

^aS.H.S. Alexander, J. Magueijo,
arXiv:hep-th/0104093

^aL. N Chang, et al., Phys. Rev. D 65, 125028 (2002).

Fermi gas equation of state

Generalized Uncertainty relation

- Number density^a is given by $\frac{m_e c^2}{K} n(\xi) = \frac{1}{\alpha^3} \left\{ \tan^{-1}(\alpha\xi) - \frac{\alpha\xi(1-\alpha^2\xi^2)}{(1+\alpha^2\xi^2)^2} \right\} = \tilde{n}$
 where $\xi = \frac{p_F}{m_e c}$ and $\alpha = m_e c \sqrt{\beta}$.
 In the limit $\xi \rightarrow \infty$, $n(\xi) \rightarrow \frac{K}{m_e c^2} \frac{\pi}{2\alpha^3}$.

- Energy density, $\varepsilon(\xi)$
 In the limit $\xi \rightarrow 0$, the energy density goes like ξ^3
 In the limit $\xi \rightarrow \infty$, the energy density approaches a constant, $\frac{\varepsilon}{K} =$

$$\frac{\pi}{2\alpha^3} (1-q) + \frac{q}{\alpha^4} \frac{(2-\alpha^2)}{(1-\alpha)^2} - q \frac{\tanh^{-1} \sqrt{1-\alpha^2}}{(1-\alpha^2)^{3/2}}$$

- Pressure, $P(\xi)$
 In the limit $\xi \rightarrow 0$, the pressure goes like ξ^4 , same as in the ideal case.
 In the limit $\xi \rightarrow \infty$, the pressure goes linear in ξ ,

$$\frac{P}{K} = \frac{\pi}{2\alpha^3} \xi - \frac{1}{\alpha^4} \frac{(2-\alpha^2)}{(1-\alpha)^2} + \frac{\tanh^{-1} \sqrt{1-\alpha^2}}{(1-\alpha^2)^{3/2}}$$

$$\text{where } K = \frac{\pi m_e^4 c^5}{h^3} \text{ and } q = \frac{m_e}{\mu_e m_u}.$$

Noncommutative geometry

- The number density^a, $n(\xi) = \frac{K}{m_e c^2} \frac{\xi^3}{3}$,
 which is same as in the case of ideal.
 But it has a maximum value corresponding to the maximum momentum cutoff ($\xi_{\max} = \frac{1}{\alpha}$) imposed by the dispersion relation, where the $\alpha = \lambda m_e c^2$
- The expressions of pressure and energy density are modified due to the modification in the dispersion relation.
- Energy density, $\varepsilon(\xi)$
 In the limit $\xi \rightarrow 0$, the energy density goes like ξ^3
 In the limit $\xi \rightarrow \xi_{\max}$, the energy density goes like $\sim -\ln(1-\alpha\xi)$
- Pressure, P
 In the limit $\xi \rightarrow 0$, the pressure goes like ξ^4 , same as in the ideal case.
 In the limit $\xi \rightarrow \xi_{\max}$, it diverges as $\sim \frac{1}{1-\alpha\xi}$.

^a A. Mathew and M. K. Nandy, RAA. **18**, 145 (2018)

^a A. Mathew and M. K. Nandy, Ann. Phys.(N.

GUP White Dwarf in Newtonian Gravity

Mass-radius relation

- To get an idea of how the effect of quantum spacetime fluctuations influence a compact star, we solve the Newtonian hydrostatic equation for white dwarfs.

- The governing equations are

$$\frac{d\xi}{dx} = -\frac{\sqrt{1+\xi^2}}{\xi} \frac{u(x)}{x^2}, \quad \frac{du}{dx} = \frac{1}{\alpha^3} \left\{ \tan^{-1}(\alpha\xi) - \frac{\alpha\xi(1-\alpha^2\xi^2)}{(1+\alpha^2\xi^2)^2} \right\} x^2.$$

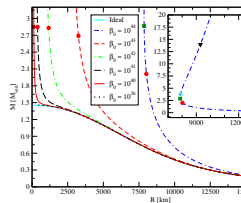
- Mass-radius curve for different strengths of α (or $\beta_0 = \beta M_P^2 c^2$) is obtained.

- We found that, in the limit $\xi \rightarrow \infty$, the star forms a core whose mass and radius scale as

$$R_{\text{core}} = \left(\frac{12\alpha^2}{\pi} \xi_c \right)^{1/2} \sqrt{1 - \frac{\xi_s}{\xi_c}}$$

$$M_{\text{core}} = \left(\frac{12\alpha^2}{\pi} \xi_c \right) \frac{1}{\sqrt{\gamma}} \left\{ \frac{\pi}{2\alpha^3} \xi_c + B(\alpha) \right\}^{1/2} \left(1 - \frac{\xi_s}{\xi_c} \right)$$

- It was observed that white dwarfs with masses far exceeding the Chandrasekhar limit can be obtained. In fact, the mass increases without bound as the central Fermi momentum ξ_c increases.
- This signifies the super stability of the star against gravitational collapse when quantum effect of spacetime fluctuations are included via GUP equation of state

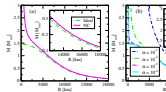


Mass-radius curve for white dwarfs for different values of β_0 . The inset display the mass-radius curve for $\beta_0 = 10^{44}$ to very large values of ξ_c . The marked points on the mass-radius curve correspond to $\alpha\xi_c = 1.00$ (solid circle), 1.25 (solid square), 1.50 (solid triangle), 3.00 (solid upright triangle).

- For large values of ξ_c the mass-radius curve deviate and it depends on the value of β_0 .

Noncommutative White dwarf in Newtonian gravity

Mass-radius relation



- A similar study was carried out using a noncommutative equation of state in the framework of Newtonian gravity.
- The figure show the mass-radius relation obtained numerically for different strengths of α (or $\alpha = \lambda m_e c^2$)
- In the high $\xi_c \rightarrow \xi_{\max}$ the mass goes as $M = \frac{M_0}{9} \left(\frac{3\alpha\xi_c}{1-\alpha\xi_c} \right)^{3/2} \zeta_R^3$, where ζ_R is the zero for the Lane-Emden function $\theta(\zeta)$ of index zero.
- This signifies the super stability of the star against gravitational collapse when quantum effect of spacetime fluctuations are included via noncommutative equation of state
- And the corresponding radius in the same limit goes as $R = \alpha \sqrt{\frac{3\alpha\xi_c}{1-\alpha\xi_c}} R_0 \zeta_R$.
- This demonstrating the possibility of excessively large values of masses and radii for very high ξ_c .
- It is important to note that behavior of mass-radius curve is similar to that observed in GUP case.

Crisis

- We observe super-stable white dwarfs in Newtonian gravity even for a very small strengths of quantum spacetime fluctuations.
- These results contradict with current observations.
- The effect of quantum gravity is inevitable, so one can not avoid it.
- We also observed that these issue can be partially resolved by incorporating Neutronization into the equation of state.
- But it was reported⁹ that there may not be a clear distinction between the gravitational core collapse and fast β -capture triggering supernova explosion in an electron degenerate core.
- Unlike Newtonian gravity in white dwarf, GTR predicts a dynamical gravitational instability¹⁰ at high central Fermi momentum.
- So, is this super-stability an artefact of Newtonian Gravity.

⁹ J.C. Wheeler, C. J. Hansen, and J.P. Cox, *Astrophys. J. Lett.* **2**, 253 (1986)

¹⁰ S. Chandrasekhar, *Phys. Rev. Lett.* **12**, 114 (1964)

General Relativistic White dwarfs

Stability Analysis

- Stability analysis is realised by allowing radial perturbation to the fluid mass^a.
- Thus, the metric interior to the fluid undergoes a radial perturbation: $ds^2 = e^{\nu+\delta\nu} dt^2 - e^{\mu+\delta\mu} dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2)$ where $\nu(r)$ and $\mu(r)$ refer to the equilibrium configuration, and $\delta\nu(r, t)$ and $\delta\mu(r, t)$ corresponds to small radial Lagrangian displacement $\zeta(r, t)$.

- Using $\zeta(r, t) = r^{-2} e^{\nu/2} \psi(r) e^{i\omega t}$, the equation governing the radial pulsation is thus obtained as

$$\frac{d}{dr} \left(U \frac{d\psi}{dr} \right) + \left(V + \frac{\omega^2}{c^2} W \right) \psi = 0$$

where

$$\begin{aligned} U(r) &= e^{(\mu+3\nu)/2} \frac{\gamma P}{r^2}, \quad V(r) = \\ &-4 \frac{e^{(\mu+3\nu)/2}}{r^3} \frac{dP}{dr} - \frac{8\pi G}{c^4} \frac{e^{3(\mu+\nu)/2}}{r^2} P(P+\varepsilon) \\ &+ \frac{e^{(\mu+3\nu)/2}}{r^2} \frac{1}{P+\varepsilon} \left(\frac{dP}{dr} \right)^2 \\ W(r) &= \frac{e^{3(\mu+\nu)/2}}{r^2} (P+\varepsilon) \end{aligned}$$

with γ the adiabatic index, $P(r)$ and $\varepsilon(r)$ are the pressure and the energy density of the fluid mass at equilibrium given by the Tolman-Oppenheimer-Volkoff equation.

- The above Sturm-Liouville equation can be derived from the action $S[\psi] =$

$$\int_0^R \left\{ U \psi'^2 - V \psi^2 - \frac{\omega^2}{c^2} W \psi^2 \right\} dr.$$

- Minimizing $S[\psi]$ with respect to ψ yields the Sturm-Liouville equation, thus providing a variational basis for determining the lowest characteristic eigenfrequency, given by

$$\frac{\omega_0^2}{c^2} = \min_{\psi(r)} \frac{\int_0^R \{ U \psi'^2 - V \psi^2 \} dr}{\int_0^R W \psi^2 dr}.$$

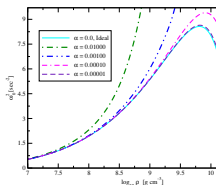
- A sufficient condition for the dynamical instability to set in is that the value of ω_0^2 be negative.

- Using suitable trial function satisfying the boundary conditions $\psi = 0$ at $r = 0$ and $\delta P = -e^{\nu/2} \frac{\gamma P}{r^2} \frac{d\psi}{dr}$ at $r = R$, we integrate the above equation to obtain frequency of the normal mode as the function of central density.

Stability Analysis

Results

- Eigen frequency for normal mode against the central density is shown for different strengths of quantum spacetime fluctuations.
- We observed that even for a large value of strength the eigenfrequencies are close to the ideal values and there exist a critical density for which the eigenfrequency vanishes.



- For strengths $\alpha > 0.0037$, no zero eigenfrequency exists and the white dwarf remains super-stable.
 - But the values of α for which the star is super stable are considered unphysical.
 - In the range $0 < \alpha < 10^{-4}$, general relativity instability yields approximately comparable critical central density as obtained for the ideal case.
 - Thus even for a high value of α , the onset density ρ_c^* for gravitational collapse is practically unaffected.
 - Our study suggests that when the inevitable effect of quantum spacetime fluctuations is included in the process, the situation practically remain indistinguishable from the ideal core collapse scenario.
- However for Higher strengths it was observed that instability sets in only for $\alpha \leq 0.0037$

Concluding Remarks and Future works

Concluding Remarks

- In both quantum-gravity scenarios—the generalized uncertainty principle and noncommutative geometry—white dwarfs are super-stable when gravity is treated within the Newtonian framework.
- We thus investigate whether this super stability prevails in general relativistic frame work.
- We find that even for a large strength of quantum spacetime fluctuations the stellar structure of relativistic white dwarf is hardly affected.
- We also observed that the onset density ρ_c^* for gravitational collapse is practically unaffected, suggesting the indistinguishability of the onset density from the ideal core collapse.
- We have already seen a similar behaviour when the GUP equation of state is employed.
- This also suggests that observing an effect of quantum spacetime fluctuations may require probing higher energy scales or phenomena such as core-collapse supernovae.

Future Works

- Mass-radius relation of relativistic white dwarfs in GUP formulation and noncommutative formulation.
- Effect of generalized uncertainty principle and noncommutative geometry in a quark-gluon plasma.
- Study the stability of such neutron core and core collapse.
- Core collapse and gravitational wave emission in the above situations.

List of Publications

- A. Mathew and M. K. Nandy, Res. Astron. Astrophys. **17**(6), 061 (2017)
- A. Mathew and M. K. Nandy, Ann. Phys.(N. Y.) **393**, 184 (2018)
- A. Mathew and M. K. Nandy, Res. Astron. Astrophys., **18**, 145 (2018)
- Dynamical instability in non-commutative white dwarfs (2018) [Submitted]
- Dynamical instability in GUP white dwarfs (2018) [In preparation]

THANK YOU